

Artificial Intelligence and Machine Learning

Project Report

Semester-IV (Batch-2022)

Crop Prediction

CHITKARA
UNIVERSITY



Submitted By: -

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1 Introduction:-

1.1 Background:

In the dynamic realm of agriculture, accurate crop prediction stands as a cornerstone for sustainable food production and economic stability. The Crop Prediction Report 2024 aims to provide a comprehensive analysis of anticipated agricultural outputs for the upcoming season. Grounded in robust data analytics and informed by historical trends, this report serves as a guiding beacon for farmers, policymakers, and stakeholders alike.

As we navigate through an era marked by climate variability and technological advancements, the significance of precise crop forecasting cannot be overstated. This report delves into various factors influencing crop yields, ranging from weather patterns and soil health to market dynamics and emerging technologies. By synthesizing this wealth of information, we strive to empower decision-makers with actionable insights to optimize resource allocation, mitigate risks, and foster resilience in agricultural systems.

Drawing upon the expertise of agronomists, meteorologists, economists, and data scientists, the Crop Prediction Report 2024 endeavors to shed light on the anticipated performance of key crops across different regions. Through meticulous analysis and forecasting models, we endeavor to unveil trends, challenges, and opportunities that lie ahead in the agricultural landscape.

In an era where food security and environmental sustainability take center stage, the importance of informed decision-making in agriculture cannot be overstated. The Crop Prediction Report 2024 serves as a beacon of knowledge, illuminating pathways towards a more resilient, productive, and equitable agricultural future.

1.2 Objectives:

Certainly! Here are some common objectives of crop prediction:

1. **Optimizing Resource Allocation:** Crop prediction helps farmers and agricultural planners allocate resources such as water, fertilizers, pesticides, and labor more efficiently by anticipating crop yield and demand.
2. **Mitigating Risks:** By forecasting potential risks such as pest infestations, diseases, adverse weather events, and market fluctuations, crop prediction enables farmers to implement preventive measures and risk management strategies.
3. **Improving Food Security:** Accurate crop prediction contributes to ensuring food security by anticipating potential shortages or surpluses, allowing for timely interventions such as imports, exports, or adjustments in planting strategies.
4. **Enhancing Economic Planning:** Crop prediction assists policymakers, governments, and agribusinesses in making informed decisions regarding investment, trade policies, subsidies, and market planning, thereby fostering economic stability and growth in the agricultural sector.
5. **Supporting Sustainable Agriculture:** By providing insights into crop performance, soil health, and climate patterns, crop prediction facilitates the adoption of sustainable agricultural practices such as precision farming, crop rotation, and soil conservation, promoting long-term environmental stewardship.
6. **Empowering Stakeholders:** By providing transparent and accessible information on crop forecasts, crop prediction empowers farmers, policymakers, researchers, and other stakeholders to actively participate in decision-making processes and contribute to the sustainable development of agriculture.

Overall, the objectives of crop prediction revolve around enhancing agricultural productivity, resilience, and sustainability while addressing the diverse needs and challenges faced by stakeholders across the agricultural value chain..

1.3 Significance:

Crop prediction holds immense significance across multiple domains of agriculture and society. Firstly, it is pivotal for ensuring food security by providing early warnings of potential crop shortages or surpluses, allowing for timely interventions to address imbalances in supply and demand. Moreover, crop prediction facilitates efficient resource management by assisting farmers in optimizing the allocation of resources such as water, fertilizers, and labor based on anticipated crop yields. Additionally, it plays a crucial role in mitigating risks associated with adverse weather conditions, pest outbreaks, and market fluctuations, empowering farmers to implement proactive strategies to protect their crops and livelihoods. Furthermore, reliable crop prediction data informs market planning and trade decisions, enabling stakeholders to anticipate market dynamics and make informed pricing and investment choices. From a policy perspective, crop prediction informs the formulation of agricultural policies and programs aimed at supporting farmers and promoting sustainable agricultural development. It also drives research and innovation by providing valuable insights into crop performance and environmental trends, contributing to the development of resilient farming practices and technologies. Ultimately, crop prediction contributes to economic stability, environmental sustainability, and resilience within agricultural systems, playing a vital role in addressing global challenges related to food production and security..

2 Problem Definition and Requirements:-

2.1 Problem Statement:

The problem statement of crop prediction typically revolves around addressing the challenges and uncertainties associated with accurately forecasting crop yields in agricultural systems. Here's a concise problem statement:

"The challenge in crop prediction lies in developing robust models and methodologies that can effectively predict future crop yields amidst the complex interplay of factors such as weather variability, soil conditions, pest and disease dynamics, and technological advancements. Achieving accurate crop predictions is essential for supporting informed decision-making by farmers, policymakers, and stakeholders across the agricultural value chain. However, existing prediction approaches often face limitations in terms of data availability, model accuracy, and scalability, hindering their ability to provide reliable forecasts in real-time

2.2 Software Requirements:

The software requirements for this project include:

Programming Language: Python

Development Environment: Jupyter Notebook

Libraries: scikit-learn, Joblib, Flask, pandas, NumPy, Matplotlib

2.3 Hardware Requirements:

The hardware requirements for this project are minimal and include a standard computer with sufficient processing power and memory to run the chosen algorithms and process large datasets.

2.4 Data Sets:

The project will utilize publicly available crop datasets, such as the Crop Recommendation, comprising a of variety of crops with suitable content of soil nutrients for training and testing the AI models.

3. Algorithms Used:-

The credit card fraud detection system employs a combination of traditional machine learning algorithms and deep learning architectures to analyse transaction data and identify fraudulent patterns effectively. The following algorithms are utilized in the project:

- **Logistic Regression:** A simple yet effective algorithm for binary classification, logistic regression is employed to model the probability of a transaction being fraudulent based on its features.
- **Decision Trees:** Decision tree-based algorithms, including Random Forests and Gradient Boosting, are utilized to capture complex nonlinear relationships and interactions between features in the transaction data.
- **Support Vector Machine:** SVM is a supervised machine learning algorithm that can be used for classification or regression tasks. SVC seeks to find the hyperplane that best separates the data points into different classes while maximizing the margin. The hyperplane is defined by a subset of the training data points called support vectors. These support vectors are the data points closest to the decision boundary or hyperplane.

These algorithms are selected based on their suitability for the credit card fraud detection task, considering factors such as interpretability, scalability, and performance. Hyperparameter tuning, cross-validation, and ensemble methods are employed to

optimize model performance, mitigate overfitting, and enhance generalization capability

4 Features :-

.NITROGEN:-

1.Data Collection -

Weather Data: Historical and current weather conditions such as temperature, rainfall, humidity, and wind speed.

Soil Data: Information about soil composition, pH levels, moisture content, and nutrient availability.

Satellite and Drone Imagery: High-resolution images to monitor crop health, growth stages, and potential issues like pests or diseases.

Agronomic Data: Details about crop types, planting and harvesting dates, crop rotation, and historical yield data.

Market Data: Information on market prices and demand for different crops.

2. Preprocessing and Data Cleaning -

Normalization: Adjusting data from different sources to a common scale.

Missing Data Handling: Filling in gaps in the data using interpolation, imputation, or removal of incomplete records.

Feature Engineering: Creating new features from raw data to improve model accuracy, such as calculating growing degree days or indices for vegetation health.

PHOSPHOROUS:-

3.Model Development-

Machine Learning Algorithms: Utilizing algorithms like linear regression, decision trees, random forests, support vector machines, and neural networks to create predictive models.

Training and Validation: Splitting data into training and validation sets to ensure models generalize well to new, unseen data.

Hyperparameter Tuning: Optimizing model parameters to improve performance.

4. Environmental and Contextual Factors

Climate Change: Incorporating trends and predictions about climate change impacts on agriculture.

Pest and Disease Outbreaks: Predicting the likelihood and potential impact of pest and disease outbreaks based on environmental conditions and historical data.

Water Availability: Considering irrigation practices and water availability for crops.

POTASSIUM:-

5. Output and Visualization

Yield Forecasts: Predicting crop yields for specific regions and time periods.

Risk Assessment: Identifying potential risks and uncertainties in crop production, such as drought or extreme weather events.

Economic Analysis: Estimating economic outcomes for different crop choices and farming practices.

Geospatial Mapping: Visualizing data on maps to show spatial variations in crop health, soil properties, and yield predictions.

6. Decision Support Systems

Advisory Services: Providing farmers with actionable recommendations on planting dates, crop selection, irrigation schedules, and pest management.

Scenario Analysis: Allowing farmers to explore different scenarios and their potential impacts on crop yields and profits.

Resource Optimization: Helping optimize the use of inputs like water, fertilizers, and pesticides to enhance productivity and sustainability.

7. Continuous Improvement

Model Updating: Regularly updating models with new data to improve accuracy and adapt to changing conditions.

Feedback Loops: Incorporating feedback from farmers and other stakeholders to refine prediction models and recommendations.

Integration with Emerging Technologies: Leveraging advancements in AI, IoT, and remote sensing to enhance data collection and predictive capabilities.

These features collectively enhance the ability to make informed decisions in agriculture, leading to increased productivity, reduced risks, and sustainable farming practices.

8. Integration with Farm Management Systems

Software Integration: Linking predictive models with farm management software for seamless data flow and actionable insights.

Automated Data Input: Automating the collection and input of data from various sources, including sensors and farm equipment, to reduce manual entry and errors.

9. Advanced Analytics

Predictive Analytics: Using historical data to forecast future trends and outcomes.

Prescriptive Analytics: Offering recommendations on actions to take based on predictive insights, such as suggesting the best planting or harvesting times.

Real-time Analytics: Providing live updates and forecasts based on real-time data from sensors, weather stations, and other sources.

5.Future Scope:-

The future scope of crop prediction is vast, driven by advancements in technology, data science, and a growing emphasis on sustainable agriculture.

1. Enhanced Accuracy and Precision

Advanced Machine Learning Models: Leveraging deep learning, reinforcement learning, and other sophisticated AI techniques to improve prediction accuracy.

Integrative Data Analysis: Combining diverse data sources, including genomic data, microbiome information, and advanced soil analytics, for more comprehensive models.

Higher Resolution Data: Using higher resolution satellite imagery and more precise ground sensors to capture detailed crop and soil conditions.

2. Real-Time and Predictive Analytics

Real-Time Monitoring: Implementing IoT devices for continuous monitoring of crop health, soil moisture, and weather conditions, allowing for immediate adjustments and interventions.

Predictive Maintenance: Forecasting equipment needs and maintenance schedules to prevent downtime and optimize farm operations.

3. Personalized Farming Solutions

Customized Recommendations: Providing highly personalized farming advice based on specific farm data, crop types, and local conditions.

Adaptive Systems: Developing systems that learn and adapt to individual farm characteristics over time, improving their recommendations with each season.

4. Integration with Smart Farming Technologies

Autonomous Machinery: Integrating crop prediction with autonomous tractors, drones, and robotic harvesters to execute precision farming techniques.

Smart Irrigation Systems: Using predictive analytics to optimize irrigation schedules, reducing water use and enhancing crop yields.

5. Climate Change Adaptation

Resilience Planning: Developing models that help farmers plan for and adapt to changing climate conditions, such as shifting growing seasons and extreme weather events.

Stress Tolerant Crops: Predicting the performance of new, stress-tolerant crop varieties under different environmental conditions.

6. Sustainability and Environmental Protection

Sustainable Practices: Enhancing models to recommend practices that minimize environmental impact, such as reducing chemical use and promoting biodiversity.

Carbon Footprint Reduction: Providing insights into practices that can help sequester carbon and reduce greenhouse gas emissions from farming.

7. Market and Economic Forecasting

Price Prediction Models: Developing advanced models to predict future market prices for crops, helping farmers make informed planting and selling decisions.

Supply Chain Optimization: Enhancing supply chain efficiency by predicting demand and optimizing the distribution of agricultural products.

8. Policy and Decision Support

Government Policy Planning: Assisting governments in policy-making by providing accurate predictions of crop yields and potential food shortages or surpluses.

Disaster Response: Improving disaster preparedness and response strategies by predicting the impacts of natural disasters on crop production.

9. Global Food Security

Global Yield Prediction Models: Creating global-scale models to predict crop yields across different regions, helping to address food security challenges.

Resource Allocation: Guiding international aid and resource allocation to regions at risk of food shortages.

10. Educational and Extension Services

Training Programs: Developing educational programs and extension services to train farmers and agronomists in using advanced prediction tools.

Community Platforms: Building online communities and platforms for sharing knowledge, experiences, and best practices among farmers globally.

11. Blockchain and Data Transparency

Blockchain Integration: Using blockchain technology to ensure data transparency, traceability, and trust in crop prediction data.

Open Data Initiatives: Promoting open data initiatives to share predictive models and data with the wider agricultural community for collaborative improvement.

12. Interdisciplinary Collaboration

Cross-Sector Partnerships: Fostering collaborations between agricultural scientists, data scientists, technologists, and policy-makers to enhance crop prediction systems.

Citizen Science Projects: Engaging farmers and communities in data collection and model validation through citizen science initiatives.

13. Ethical and Equitable Access

Equitable Access: Ensuring smallholder and resource-poor farmers have access to advanced crop prediction tools and technologies.

Ethical Use of Data: Promoting ethical standards in the collection, use, and sharing of agricultural data.

- The future of crop prediction is poised to revolutionize agriculture by making it more efficient, sustainable, and resilient to the challenges of the 21st century.

4 Snapshots:-

Jupyter Crop_prediction Last Checkpoint: 14 minutes ago (unsaved changes)

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel)

```
In [1]: 1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 import joblib
```

```
In [2]: 1 cr=pd.read_csv('cp.csv')      #here we upload the csv file to pandas dataframe
2 cr.head()                        #head function will give first 5 rows of dataset
```

```
Out[2]:
```

	N	P	K	temperature	humidity	ph	rainfall	label
0	90	42	43	20.879744	82.002744	6.502965	202.935536	rice
1	85	58	41	21.770462	80.319644	7.038096	226.655537	rice
2	60	55	44	23.004459	82.320763	7.840207	263.964248	rice
3	74	35	40	26.491096	80.158363	6.980401	242.864034	rice
4	78	42	42	20.130175	81.604873	7.628473	262.717340	rice

```
In [3]: 1 cr.columns
```

```
Out[3]: Index(['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall', 'label'], dtype='object')
```

```
In [4]: 1 cr['label'].value_counts()      #count of particular label
```

```
Out[4]: label
rice      100
maize     100
jute      100
cotton    100
coconut   100
papaya    100
orange    100
apple     100
muskmelon 100
watermelon 100
grapes    100
mango     100
banana    100
pomegranate 100
```

Jupyter Crop_prediction Last Checkpoint: 15 minutes ago (unsaved changes)

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel)

```
In [8]: 1 X=cr.drop(['label'],axis=1)
2 y=cr['label']
```

```
In [9]: 1 sns.heatmap(X.corr(),annot=True)
```

```
Out[9]: <Axes: >
```

```
In [10]: 1 acc=[]
2 model=[]
```

```
In [11]: 1 from sklearn.model_selection import train_test_split
```

```
In [12]: 1 X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)
```

Jupyter Crop_prediction Last Checkpoint: 15 minutes ago (autosaved) Logout

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel)

```
In [11]: 1 from sklearn.model_selection import train_test_split

In [12]: 1 X_train,X_test,Y_train,Y_test=train_test_split(X,y,test_size=0.2,random_state=42)

In [13]: 1 X_train.shape,X_test.shape,Y_train.shape,Y_test.shape
Out[13]: ((1760, 7), (440, 7), (1760,), (440,))

In [14]: 1 from sklearn.linear_model import LogisticRegression

In [15]: 1 import warnings
2 warnings.filterwarnings('ignore')

In [16]: 1 log_reg=LogisticRegression()
2 log_reg.fit(X_train,Y_train)
Out[16]: LogisticRegression
LogisticRegression()

In [17]: 1 y_pred=log_reg.predict(X_test)

In [18]: 1 from sklearn.metrics import accuracy_score

In [19]: 1 print("accuracy_score",accuracy_score(Y_test,y_pred)*100)
accuracy_score 94.54545454545455

In [20]: 1 from sklearn.svm import SVC

In [21]: 1 svc_clf=SVC()
2 svc_clf.fit(X_train,Y_train)
Out[21]: SVC
SVC()

In [22]: 1 y_pred=svc_clf.predict(X_test)
```

Jupyter Crop_prediction Last Checkpoint: 16 minutes ago (autosaved) Logout

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel)

```
In [22]: 1 y_pred=svc_clf.predict(X_test)

In [23]: 1 print("accuracy_score",accuracy_score(Y_test,y_pred)*100)
accuracy_score 96.13636363636363

In [24]: 1 from sklearn.linear_model import LogisticRegression
2 from sklearn.svm import SVC
3 from sklearn.tree import DecisionTreeClassifier

In [25]: 1 clfs=[LogisticRegression(),
2 SVC(kernel='rbf'),
3 DecisionTreeClassifier(min_samples_leaf=10,random_state=42),
4 DecisionTreeClassifier(max_depth=10)]

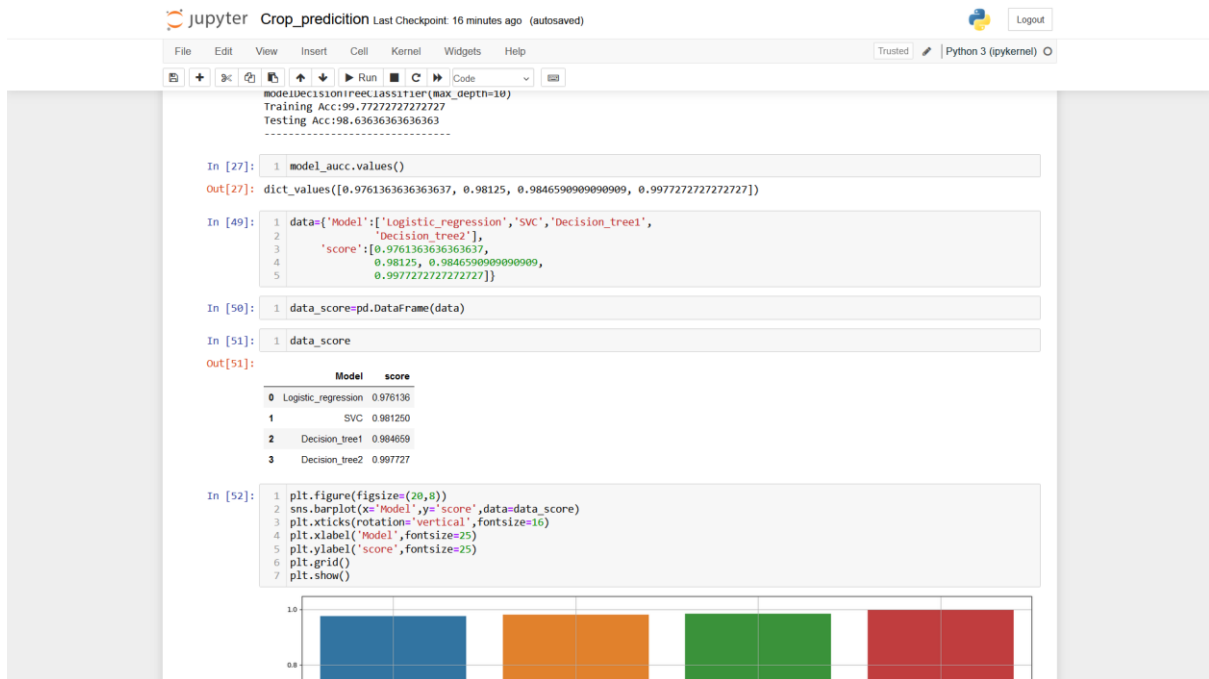
In [26]: 1 model_aucc={}
2 for clf in clfs:
3     print(f'model{clf}')
4     clf.fit(X_train,Y_train)
5
6     print(f'Training Acc: {(clf.score(X_train,Y_train)*100)}')
7     print(f'Testing Acc: {(clf.score(X_test,Y_test)*100)}')
8     model_aucc.update({clf:clf.score(X_train,Y_train)})
9     print("-----")

modelLogisticRegression()
Training Acc:97.61363636363637
Testing Acc:94.54545454545455

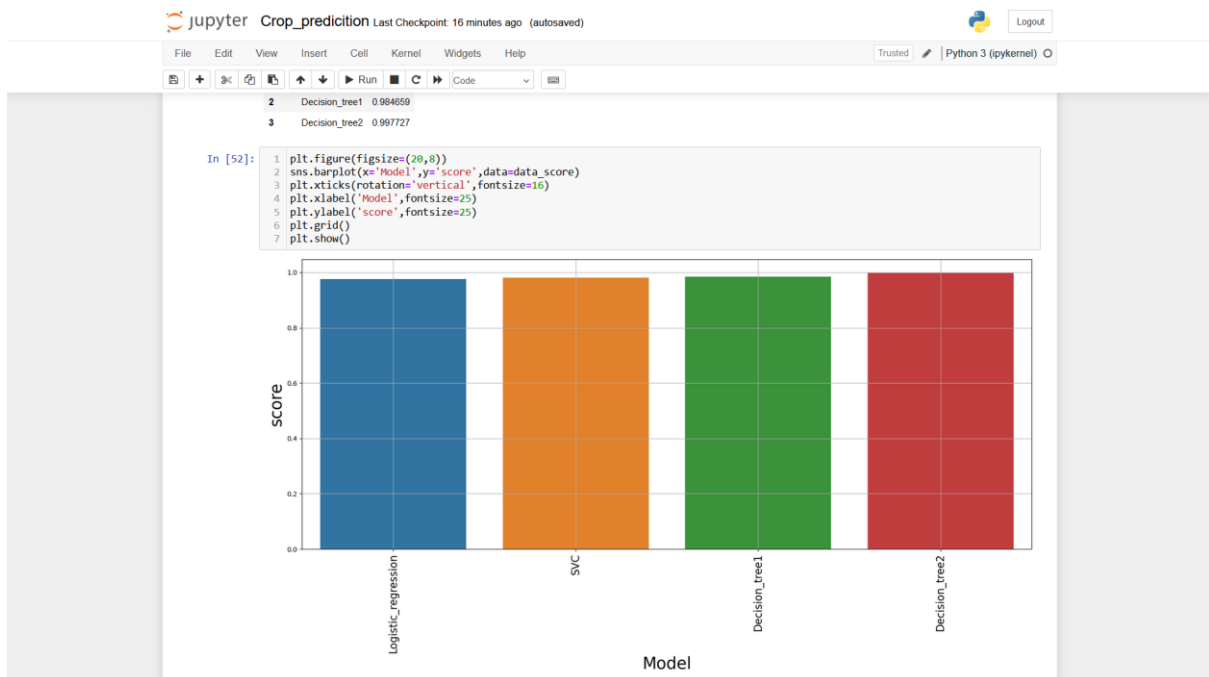
modelSVC()
Training Acc:98.125
Testing Acc:96.13636363636363

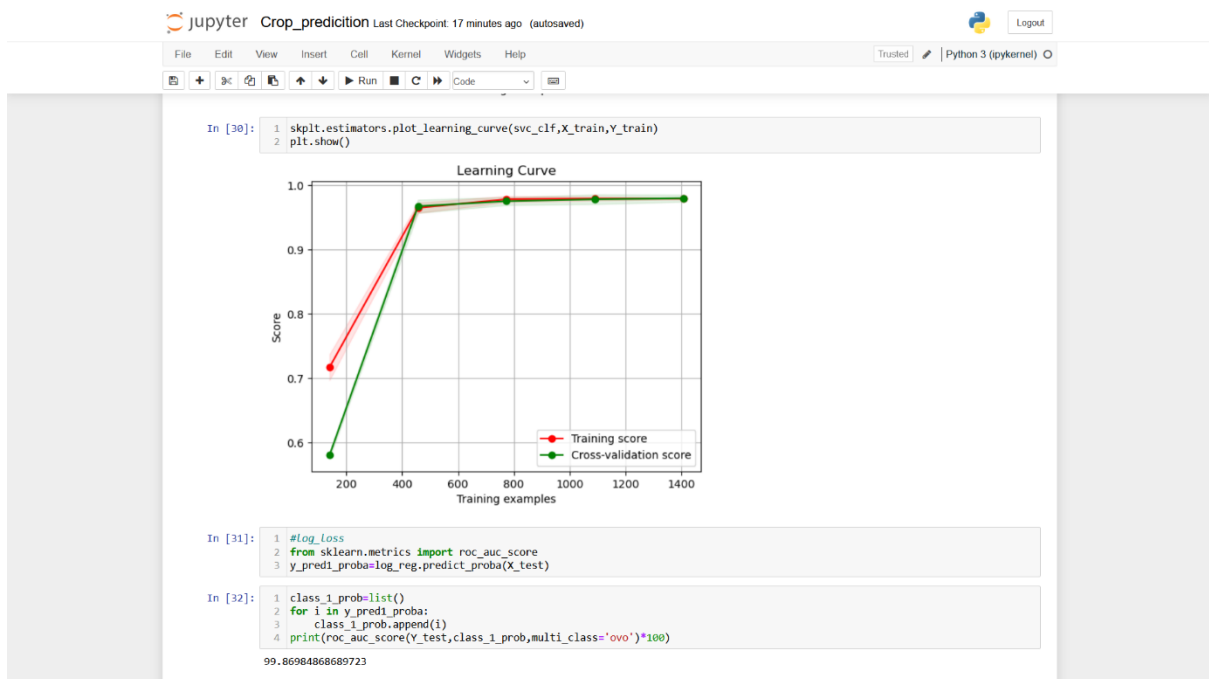
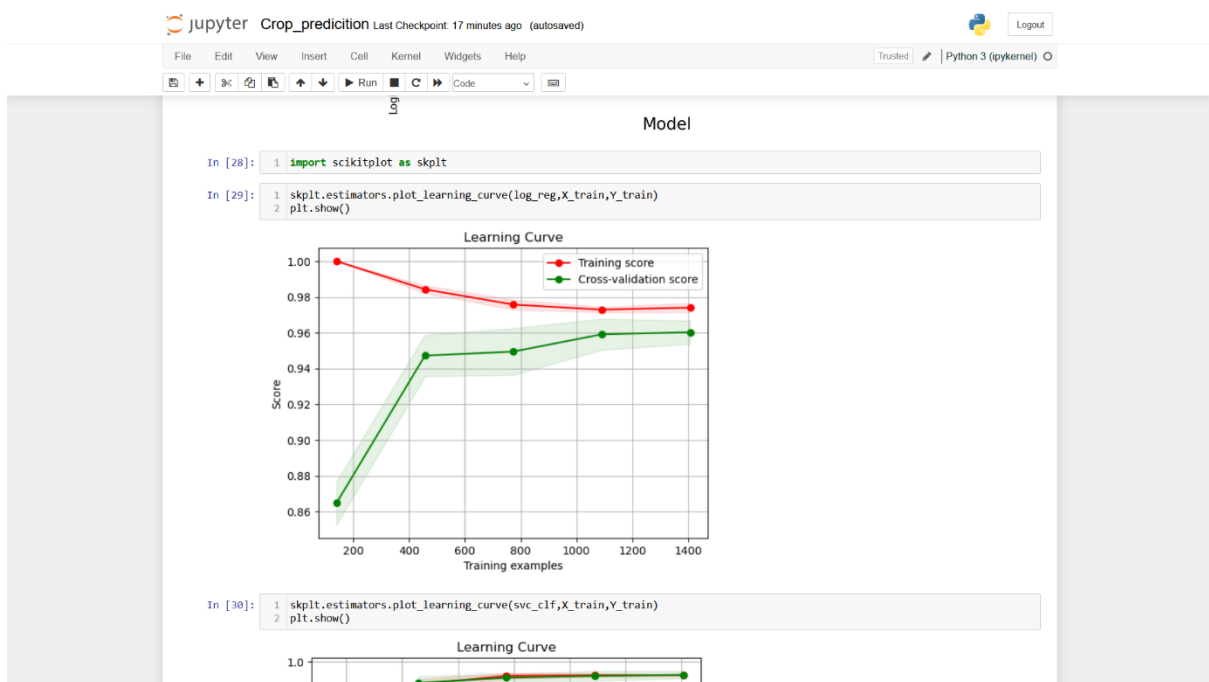
modelDecisionTreeClassifier(min_samples_leaf=10, random_state=42)
Training Acc:98.46590909090909
Testing Acc:97.27272727272728

modelDecisionTreeClassifier(max_depth=10)
Training Acc:99.77272727272727
Testing Acc:98.63636363636363
-----
```



S





```
jupyter Crop_prediction Last Checkpoint: 17 minutes ago (autosaved) Python 3 (ipykernel)

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel)

In [31]: 1 #log_loss
         2 from sklearn.metrics import roc_auc_score
         3 y_predi_proba=log_reg.predict_proba(X_test)

In [32]: 1 class_1_prob=list()
         2 for i in y_predi_proba:
         3     class_1_prob.append(i)
         4 print(roc_auc_score(Y_test,class_1_prob,multi_class='ovo')*100)

99.86984868689723

In [33]: 1 pip install joblib

Requirement already satisfied: joblib in c:\users\hites\anaconda3\lib\site-packages (1.2.0)
Note: you may need to restart the kernel to use updated packages.

In [ ]: 1

In [34]: 1 import joblib

In [ ]: 1

In [35]: 1 file_name = "crop_model"

In [36]: 1 joblib.dump(log_reg,file_name)

Out[36]: ['crop_model']

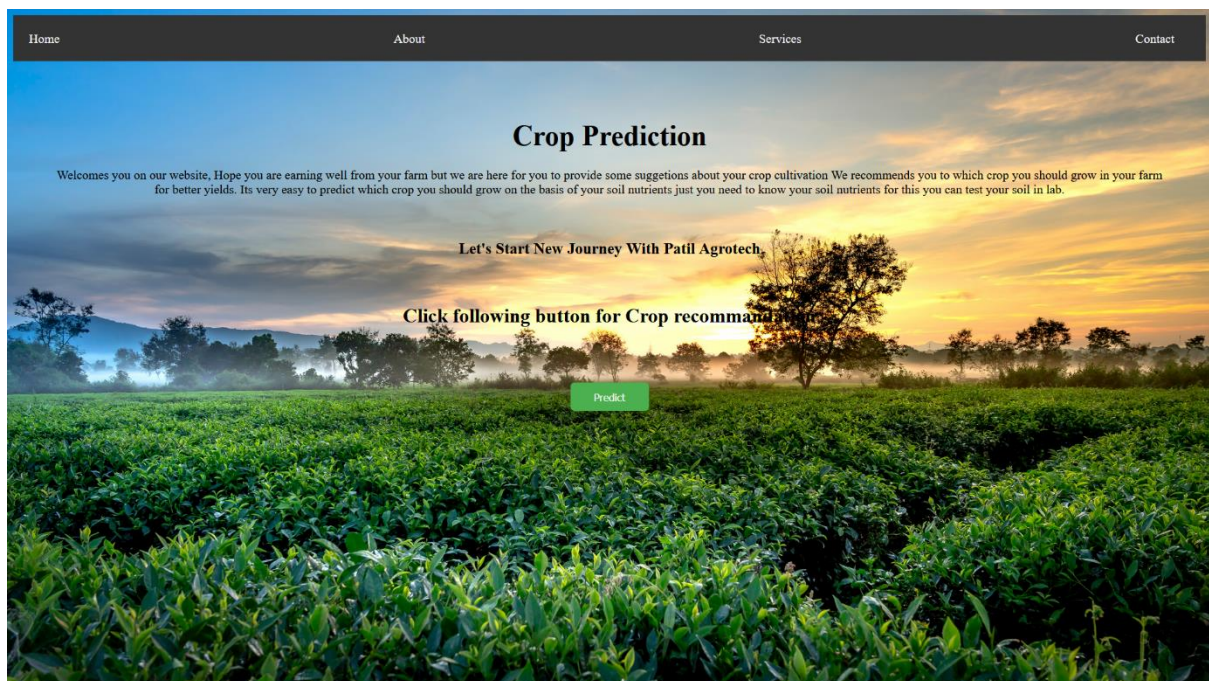
In [37]: 1 app = joblib.load('crop_model')

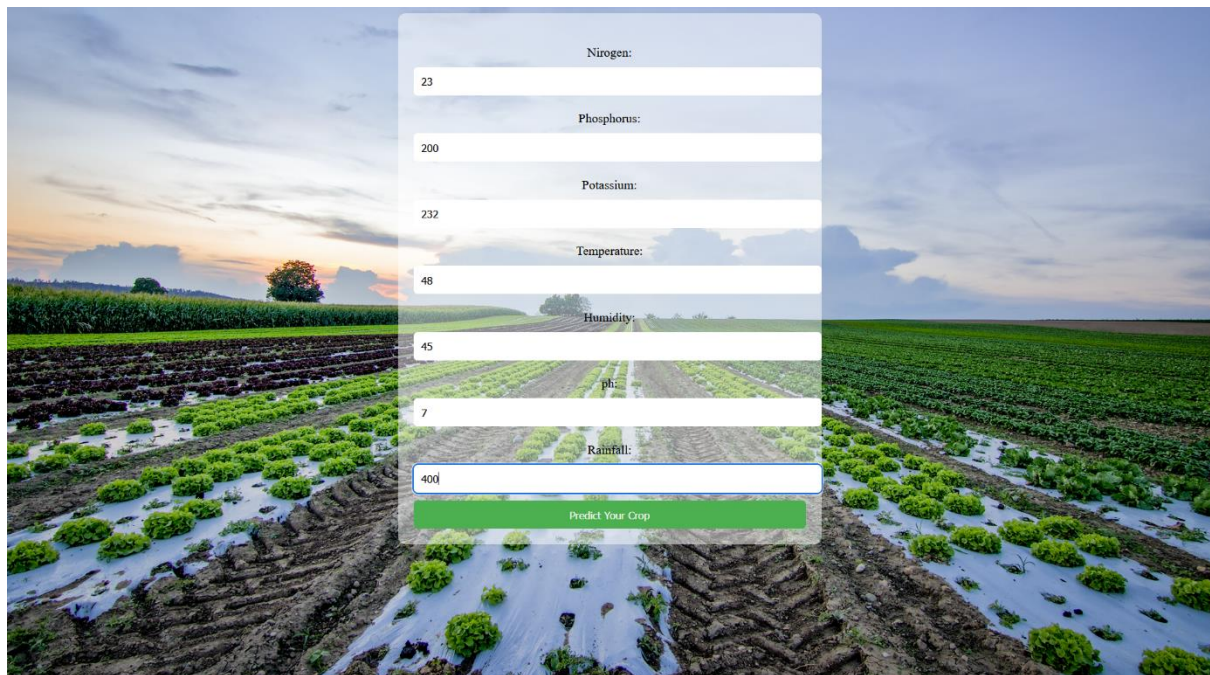
In [38]: 1 arr = [[90,42,43,20,879744,82.002744,6.502985,202.935536]]
         2 acc = app.predict(arr)

In [39]: 1 acc

Out[39]: array(['rice'], dtype=object)

In [40]: 1 import pickle
```





Nitrogen:
23

Phosphorus:
200

Potassium:
232

Temperature:
48

Humidity:
45

pH:
7

Rainfall:
400

Predict Your Crop

